

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

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We separate the sign and positive-amplitude parts of a real many-body wave function, $\psi(x) = s(x)a(x)$ with $a(x) = |\psi(x)| > 0$, in the frustrated square-lattice spin-1/2 J_1 - J_2 Heisenberg model. For a fixed amplitude, one shifted Krylov step gives $s'(x) = s(x) \text{sgn}[T - r_s(x)]$, where $r_s(x) = (H\psi)(x)/\psi(x)$. At $J_2/J_1 = 0.5$ this reduces the exact 4×4 probability weight on incorrectly signed configurations from 1.27×10^{-2} to 3.41×10^{-4} . On 6×6 , using exactly the same neural-network amplitude, the reconstructed signs give the same energy as the original neural-network signs within sampling uncertainty. Conversely, lattice fixed-node projection improves the positive amplitude while keeping the signs fixed; an exact 4×4 fixed-node/Krylov loop reaches the exact ground-state signs. The reconstructed signs remain useful on 8×8 . The remaining scaling problem is to infer from fixed-node Monte Carlo walkers a positive amplitude function that can be evaluated on arbitrary configurations. Maximum likelihood targets $a_{\text{new}}^2 \propto a\phi_{\text{FN}}$, and stochastic reconfiguration produces an update that generalizes between independent walker samples.

I. INTRODUCTION

The Monte Carlo sign problem is severe in frustrated spin systems [1, 2]. We use the square-lattice J_1 - J_2 Heisenberg model as a benchmark. At $J_2 = 0$ its ground-state signs obey the Marshall rule [3]; near $J_2/J_1 = 1/2$ that rule fails and accurate variational states become nontrivial [4, 5].

We ask whether signs and positive amplitudes can be improved separately. For a real wave function we write

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) = |\psi(x)| > 0. \quad (1)$$

Given a , a single Krylov step updates the signs. Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. Our calculations first test these two operations separately and then close the feedback loop.

II. MODEL AND METHOD

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic $L \times L$ lattices with $J_1 = 1$ in the $S^z = 0$ sector. For a spin configuration x , the Marshall rule is $s_M(x) = (-1)^{N_{\downarrow}^A(x)}$ up to an overall sign, where $N_{\downarrow}^A$ counts down spins on one checkerboard sublattice.

A. One-step sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (3)$$

Because

$$[(T - H)\psi](x) = [T - r_s(x)]\psi(x), \quad (4)$$

the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \text{sgn}[T - r_s(x)]. \quad (5)$$

Only this sign is kept; the Krylov amplitude factor $|T - r_s(x)|$ is discarded. Thus the method is not repeated Lanczos multiplication. For exact systems, T is chosen to minimize the fixed-amplitude variational energy. For sampled systems, T is obtained from a two-cluster split of the $r_s(x)$ values and therefore does not use target sign labels.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we fix the irrelevant overall sign and report directly the wrong-sign probability weight

$$w_s = \sum_{x: s(x) \neq s_0(x)} a_0(x)^2. \quad (6)$$

B. Fixed-node amplitude update

For a trial wave function ψ_T , an off-diagonal matrix element connecting x and y is sign violating when $H_{xy}\psi_T(y)/\psi_T(x) > 0$. Lattice fixed node removes such off-diagonal terms and adds their contribution to the diagonal [6]. After importance sampling, the resulting Hamiltonian is sign free and keeps the signs of ψ_T fixed.

Let $\phi_{\text{FN}}(x) > 0$ be the positive ground-state amplitude obtained from this fixed-node Hamiltonian. In an exact small-system calculation, ϕ_{FN} can be inserted directly into Eq. (5). In a walker calculation, however, the stationary sampled distribution is

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

If a positive model is trained by maximum likelihood with $p_\theta(x) \propto a_\theta(x)^2$, the exact maximum-likelihood target is

$$a_{\text{new}}^2(x) \propto a(x)\phi_{\text{FN}}(x). \quad (8)$$

Since $a_{\text{new}}(x) > 0$, this is equivalent to taking the geometric mean of the current and fixed-node amplitudes.

The 4×4 exact calculations use the full $S^z = 0$ Hilbert space of 12870 states. For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. In the sign-only benchmarks, its positive amplitude $|\psi_{\text{ViT}}(x)|$ is held fixed. The 8×8 fixed-node calculations use a walker population $M = 128$ and projection length $\beta = 1.2$.

III. ONE-STEP SIGN RECONSTRUCTION

Figure 1(a,b) uses the exact 4×4 positive ground-state amplitude. At $J_2/J_1 = 0.5$, Marshall signs place 1.27×10^{-2} of the exact ground-state probability on configurations with the wrong sign. One Krylov update reduces this to 3.41×10^{-4} and lowers the fixed-amplitude energy error per site from 1.16×10^{-2} to 5.51×10^{-4} .

The 6×6 test no longer uses exact signs. With the same ViT positive amplitude for all three states,

$$\begin{aligned} e_{\text{Marshall}} &= -0.4981534, \\ e_{\text{Krylov}} &= -0.5033337, \\ e_{\text{ViT}} &= -0.5034504. \end{aligned} \quad (9)$$

The paired difference $e_{\text{Krylov}} - e_{\text{ViT}} = 1.17(287) \times 10^{-4}$ is unresolved, whereas Krylov improves over Marshall by 4.79×10^{-3} per site. The energy gain therefore comes from the signs, not from changing the amplitude.

The mechanism is visible in the Marshall local energy,

$$r_M(x) = D(x) + F_{J_1}(x) + F_{J_2}(x), \quad (10)$$

where D is the diagonal contribution and F_{J_1}, F_{J_2} are the two off-diagonal exchange contributions. The useful signal is mainly their competition. On two independent 6×6 samples, the fraction of $|\psi_{\text{ViT}}|^2$ weight with the wrong sign falls from 0.01635 to 8.08×10^{-4} and from 0.01811 to 8.02×10^{-4} , respectively.

IV. CLOSING THE LOOP EXACTLY

We target $J_2/J_1 = 0.5$ but initialize with the exact positive ground-state amplitude at $J_2 = 0$ and Marshall signs. At each iteration we solve the current fixed-node problem exactly, replace the positive amplitude by the fixed-node ground-state amplitude, and recompute Eq. (5). The exact $J_2/J_1 = 0.5$ ground state is used only to evaluate the plotted errors.

Initially $F_a = 0.7132$ and the wrong-sign weight is $w_s = 1.27 \times 10^{-2}$. After two iterations $w_s = 2.03 \times 10^{-3}$; after twenty, $w_s = 8.85 \times 10^{-7}$; at iteration 100 it is zero to numerical precision. The final amplitude fidelity is 0.9999946, and the energy of the current trial state is 2.60×10^{-5} above the exact ground state.

The plateaus in Fig. 2(a) have a simple interpretation: fixed node can improve the amplitude for several iterations without changing any sign. A new sign pattern appears only when the improved amplitude moves the local-energy threshold across additional configurations.

V. SCALING TO 8×8

At $M = 128$, the mean total fixed-node energies from two independent walker populations are

$$E_{\text{Krylov}} = -31.88542, \quad E_{\text{Marshall}} = -31.67748. \quad (11)$$

Their difference is -0.20795 . With only two populations, their spread is used only as a reproducibility scale.

Using the Krylov-reconstructed signs gives $E/N = -0.498210$, compared with -0.494961 when the same fixed-node calculation uses Marshall signs. Figure 3 shows that it lies in the same energy range as high-accuracy DMRG/FAMPS and neural benchmarks on the same geometry. We use this only as a scale comparison, not as a ranking between methods.

Thus the reconstructed signs remain useful on 8×8 . The remaining problem is to infer from fixed-node walkers a positive amplitude function that can be evaluated on configurations not present in the walker sample.

VI. LEARNING THE AMPLITUDE FROM WALKERS

Equation (8) can be tested exactly on 4×4 . Starting from the exact $J_2/J_1 = 0.5$ ground-state amplitudes but assigning Marshall signs, one exact amplitude update from Eq. (8) followed by the Krylov sign update changes the energy from -8.27291 to -8.44579 , close to $E_0 = -8.45792$. A deliberately poor start in which all positive amplitudes are equal initially moves to higher energy, but later iterations recover [Fig. 4(a)].

At 8×8 , let g_1 and g_2 be maximum-likelihood gradients estimated from two independent fixed-node walker samples. Their Euclidean cosine is

$$\frac{g_1 \cdot g_2}{\|g_1\| \|g_2\|} = -0.853, \quad (12)$$

so an ordinary gradient step fitted to one sample does not generalize to the other.

We therefore use stochastic reconfiguration (SR) [8],

$$\delta\theta = (S + \lambda I)^{-1} g_1, \quad (13)$$

where S is the empirical Fisher matrix of the amplitude model and λ is a diagonal regularization. For $\lambda = 1$, the directional derivative on the independent second walker sample becomes $g_2^\top \delta\theta = +3.85$. The finite step-size scan in Fig. 4(b) remains positive and peaks near $\eta = 0.01$, where the log-likelihood change on the independent sample is $+0.0156$ and the reweighting effective sample size is 3236/4096.

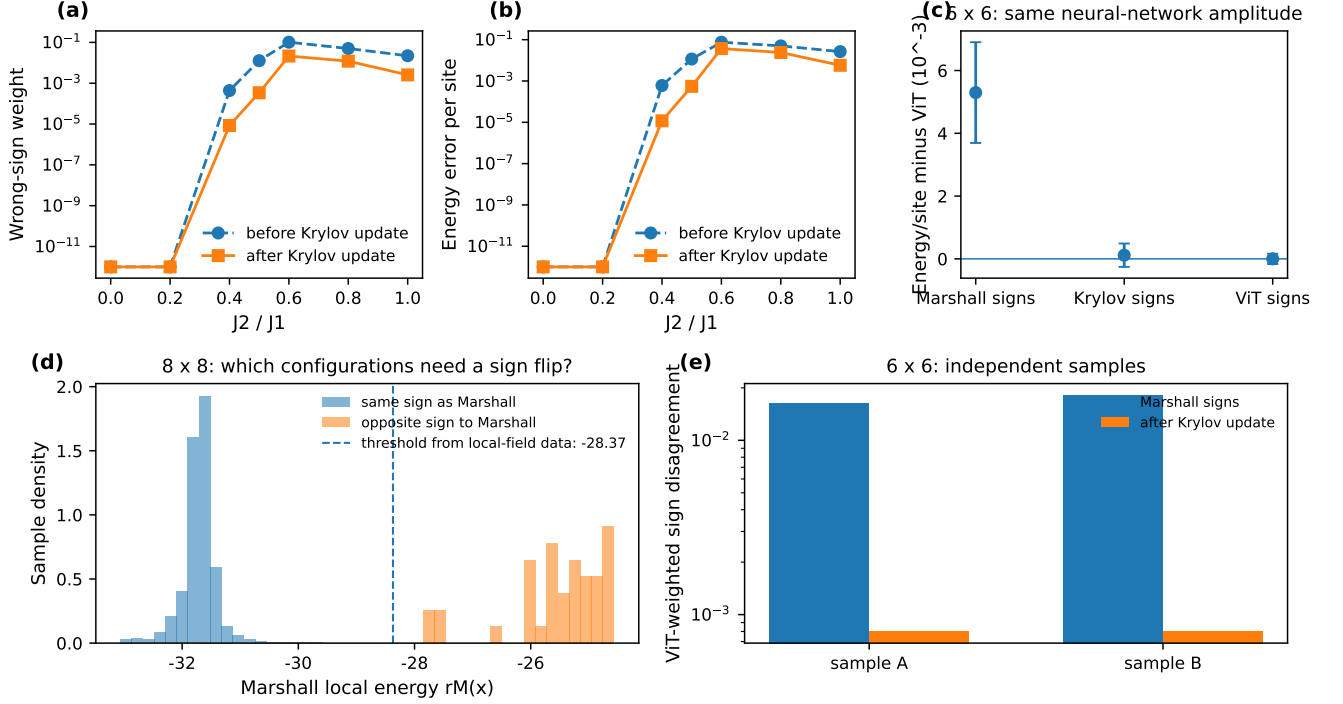


FIG. 1. Sign reconstruction. (a,b) Exact 4×4 wrong-sign weight w_s and fixed-amplitude energy error. The initial sign rule is Marshall for $J_2/J_1 \leq 0.6$ and a stripe rule for $J_2/J_1 \geq 0.8$; the stripe rule uses alternating columns instead of the checkerboard sublattices of the Marshall rule. The positive amplitude is always the exact ground-state amplitude. (c) 6×6 energies obtained with the same ViT positive amplitude and three different sign patterns; error bars are between-chain standard errors. (d) 8×8 distributions of the Marshall local energy $r_M(x)$. ViT signs are used only afterward to identify configurations that agree or disagree with Marshall; the threshold is obtained from the r_M values alone. (e) Fraction of the ViT probability weight $|\psi_{\text{ViT}}|^2$ whose sign disagrees with the ViT sign, before and after the Krylov update, on two independent samples. Exact zeros in panels (a,b) are displayed at 10^{-12} only to make them visible on the logarithmic axes.

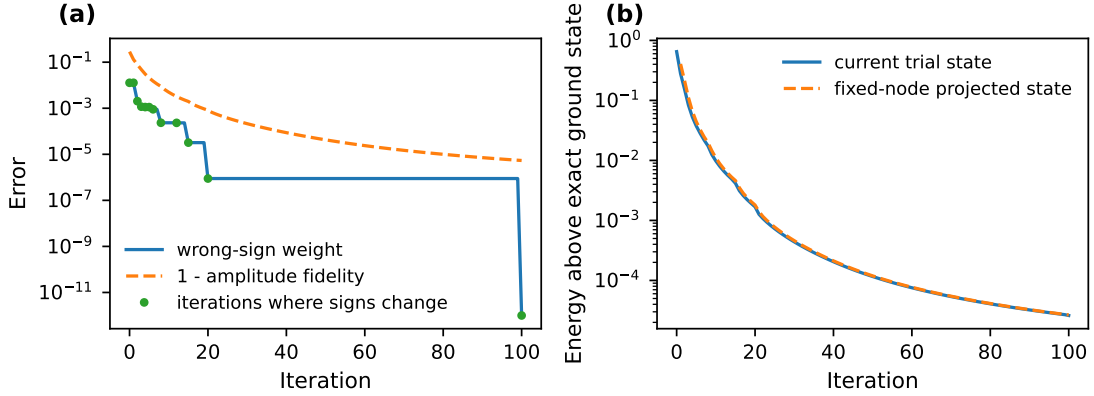


FIG. 2. Exact 4×4 fixed-node/Krylov loop at $J_2/J_1 = 0.5$. (a) Wrong-sign weight w_s and amplitude error $1 - F_a$. For normalized positive amplitudes, $F_a = |\langle a|a_0 \rangle|^2$. Markers indicate iterations at which the sign pattern changes. The exact zero reached at the end is displayed at 10^{-12} on the logarithmic axis. (b) Energy above the exact ground state for the current trial state and for the fixed-node projected state.

This shows that one learned amplitude update generalizes between independent walker samples. It does not yet establish convergence of repeated 8×8 updates; the fixed-node energy remains the acceptance test.

VII. DISCUSSION AND OUTLOOK

The data separate the two tasks. With a good positive amplitude, the dominant frustrated sign correction is

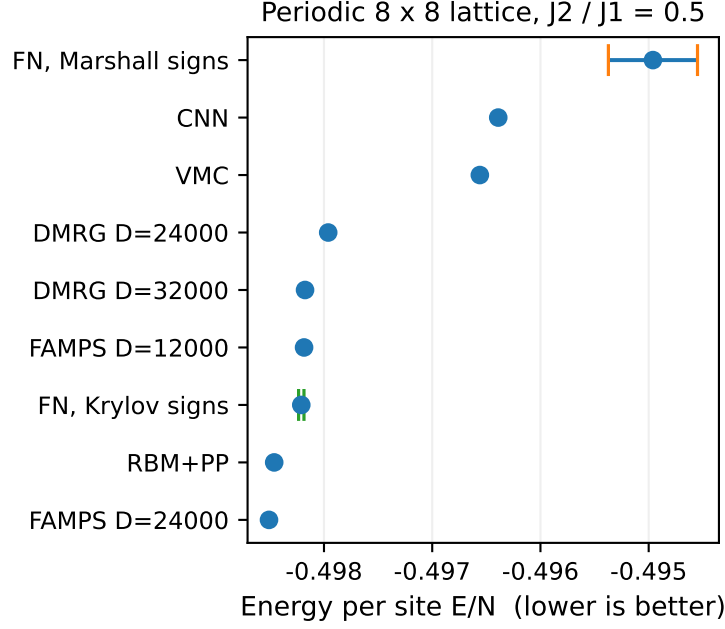


FIG. 3. Energy per site on the periodic 8×8 lattice at $J_2/J_1 = 0.5$. “FN” denotes lattice fixed node. The two FN entries use Marshall signs or the one-step Krylov-reconstructed signs. Short ticks show the two independent walker populations. Literature values are from Ref. [7].

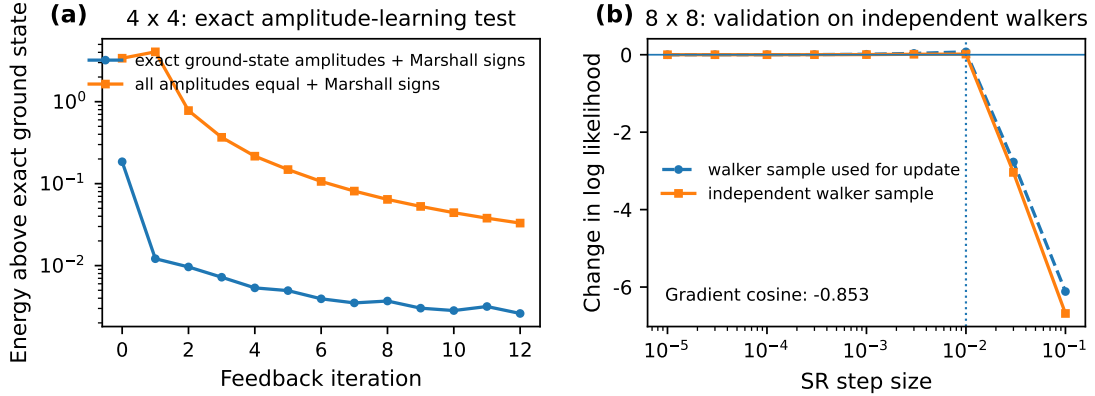


FIG. 4. Learning the positive amplitude. (a) Exact 4×4 test of Eq. (8). “Exact amplitudes” means the exact $J_2/J_1 = 0.5$ ground-state amplitudes; “equal amplitudes” means the same positive value $a(x) = \text{const.}$ for every basis state, followed by normalization. Both start with Marshall signs. (b) Stochastic-reconfiguration (SR) step-size scan on 8×8 : the update is fitted to one fixed-node walker sample and evaluated on an independent sample.

captured by a threshold in the local energy. Fixed node solves the complementary positive-amplitude problem at fixed signs, and the exact 4×4 loop shows that the two updates can close.

The unresolved large-system problem is learning that positive amplitude from projector walkers. SR makes one 8×8 update generalize between independent walker samples, but repeated updates still require fixed-node energy validation. The method also requires a useful initial amplitude and should not be confused with repeated bare Krylov iteration; the direct second step is worse on 6×6 .

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second unsupervised threshold changes only 0.8–2.0% of the $|\psi_{\text{VIT}}|^2$ probability weight but raises the fixed-amplitude energy by 0.65–0.96 in total energy. The feedback construction therefore restores or re-solves the amplitude before another sign update.

Appendix B: Statistical convention

Exact-diagonalization data are deterministic up to eigensolver precision. For the 6×6 fixed-amplitude com-

parison we use correlation-robust between-chain standard errors. For 8×8 fixed-node data with two independent walker populations, error bars denote half the population difference and are treated only as reproducibility scales.

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- [1] P. Henelius and A. W. Sandvik, Sign problem in monte carlo simulations of frustrated quantum spin systems, *Phys. Rev. B* **62**, 1102 (2000).
 - [2] M. Troyer and U.-J. Wiese, Computational complexity and fundamental limitations to fermionic quantum monte carlo simulations, *Phys. Rev. Lett.* **94**, 170201 (2005).
 - [3] W. Marshall, Antiferromagnetism, *Proc. R. Soc. Lond. A* **232**, 48 (1955).
 - [4] K. Choo, T. Neupert, and G. Carleo, Two-dimensional frustrated j_1 - j_2 model studied with neural network quantum states, *Phys. Rev. B* **100**, 125124 (2019).
 - [5] A. Chen and M. Heyl, Empowering deep neural quantum states through efficient optimization, *Nat. Phys.* **20**, 1476 (2024).
 - [6] H. J. M. van Bommel, D. F. B. ten Haaf, W. van Saarloos, J. M. J. van Leeuwen, and G. An, Fixed-node quantum monte carlo method for lattice fermions, *Phys. Rev. Lett.* **72**, 2442 (1994).
 - [7] X. Qian and M. Qin, Augmenting density matrix renormalization group with disentanglers, *Chin. Phys. Lett.* **40**, 057102 (2023).
 - [8] S. Sorella, Generalized lanczos algorithm for variational quantum monte carlo, *Phys. Rev. B* **64**, 024512 (2001).